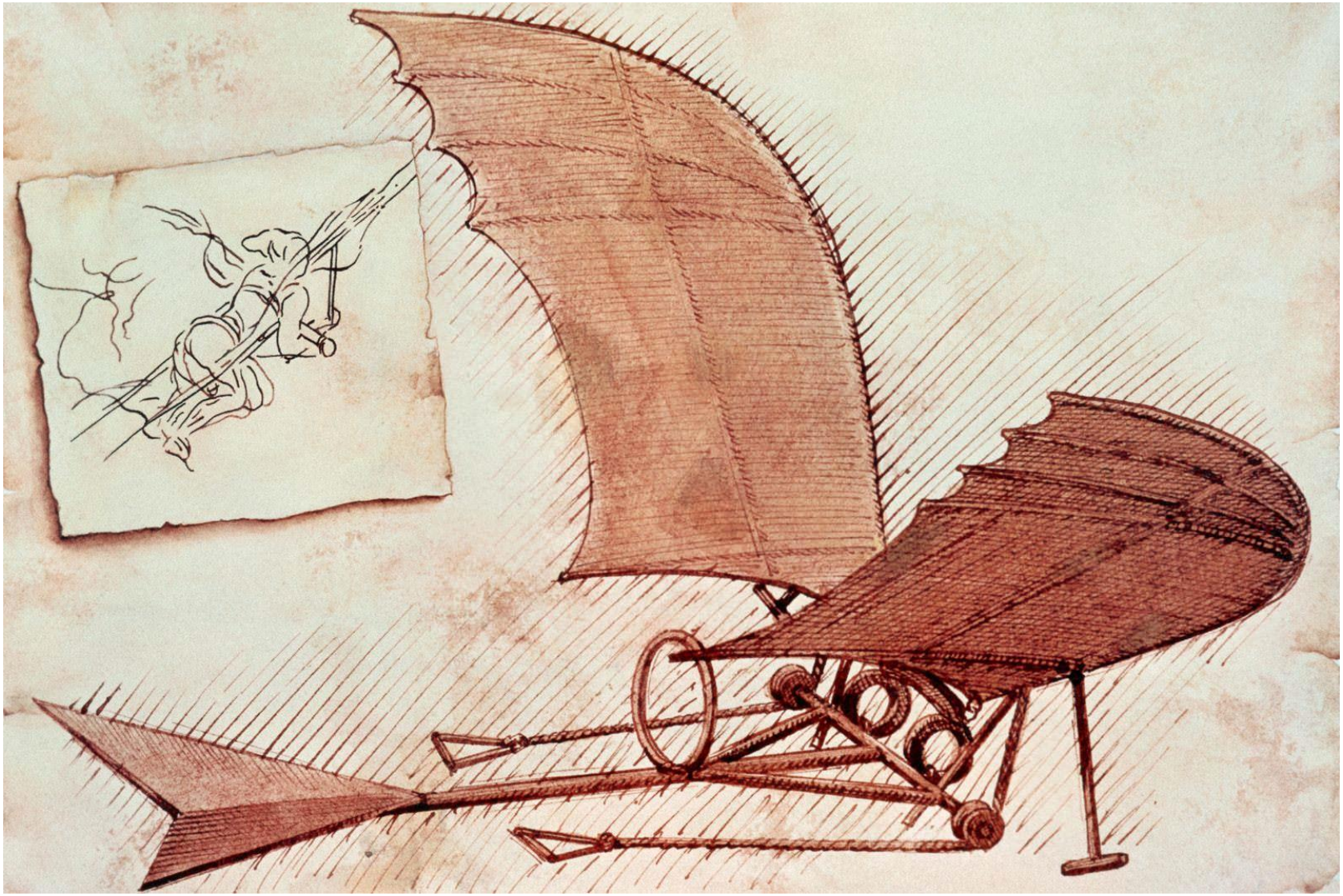
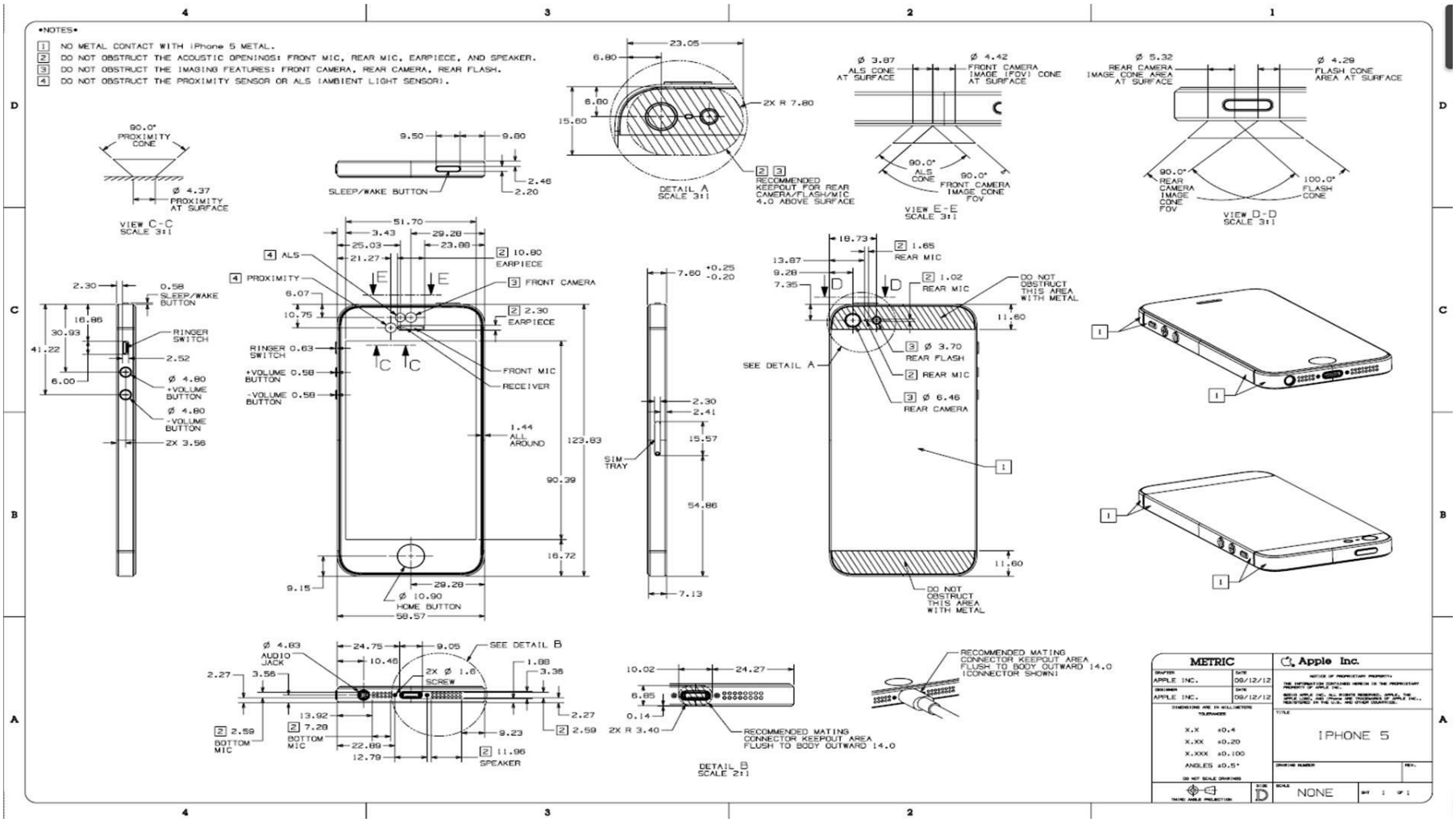
ME1480 Engineering Drawing

Bright Idea (Freehand Sketch)

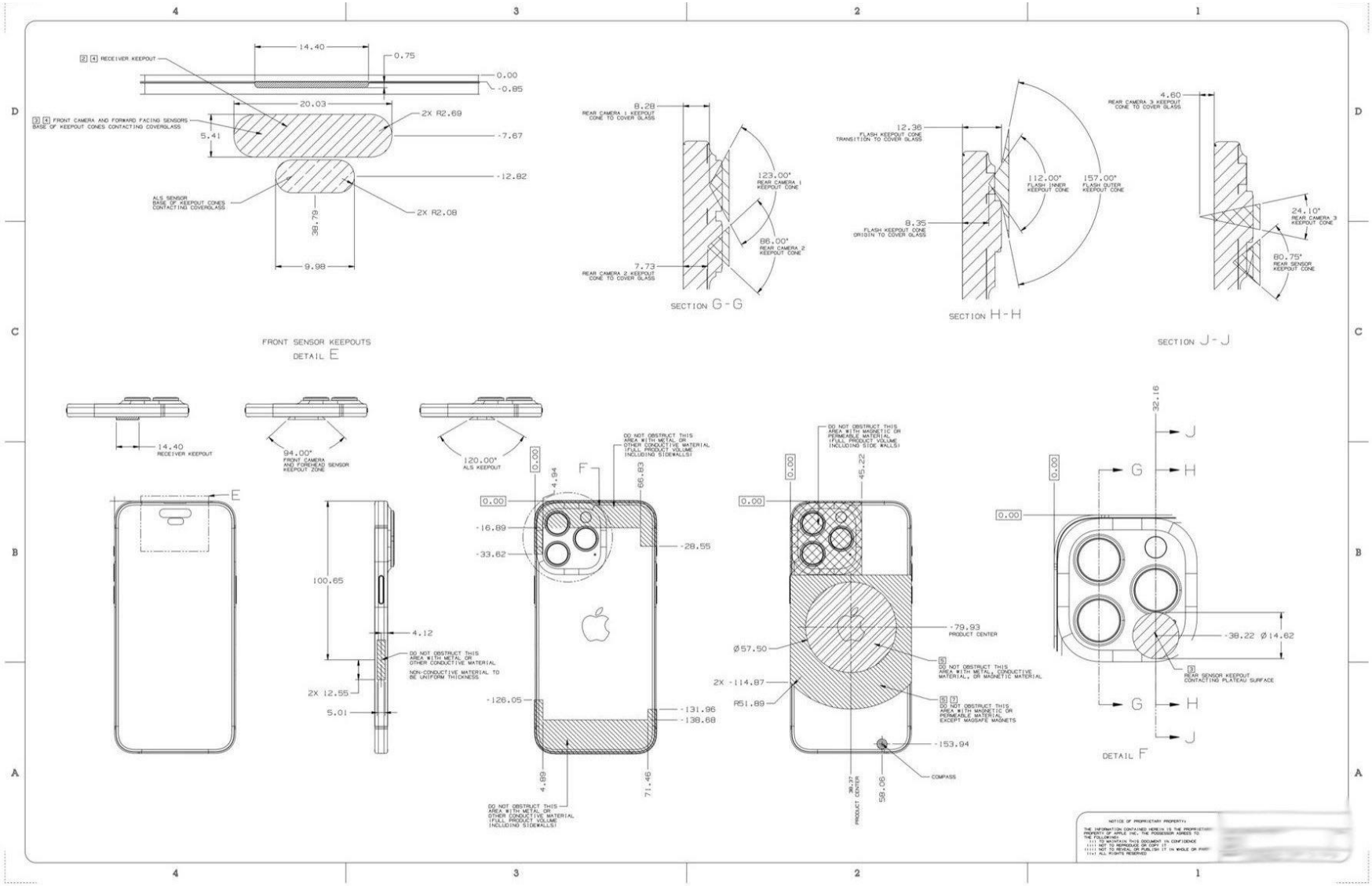


Flying Machine Leonardo Da Vinci

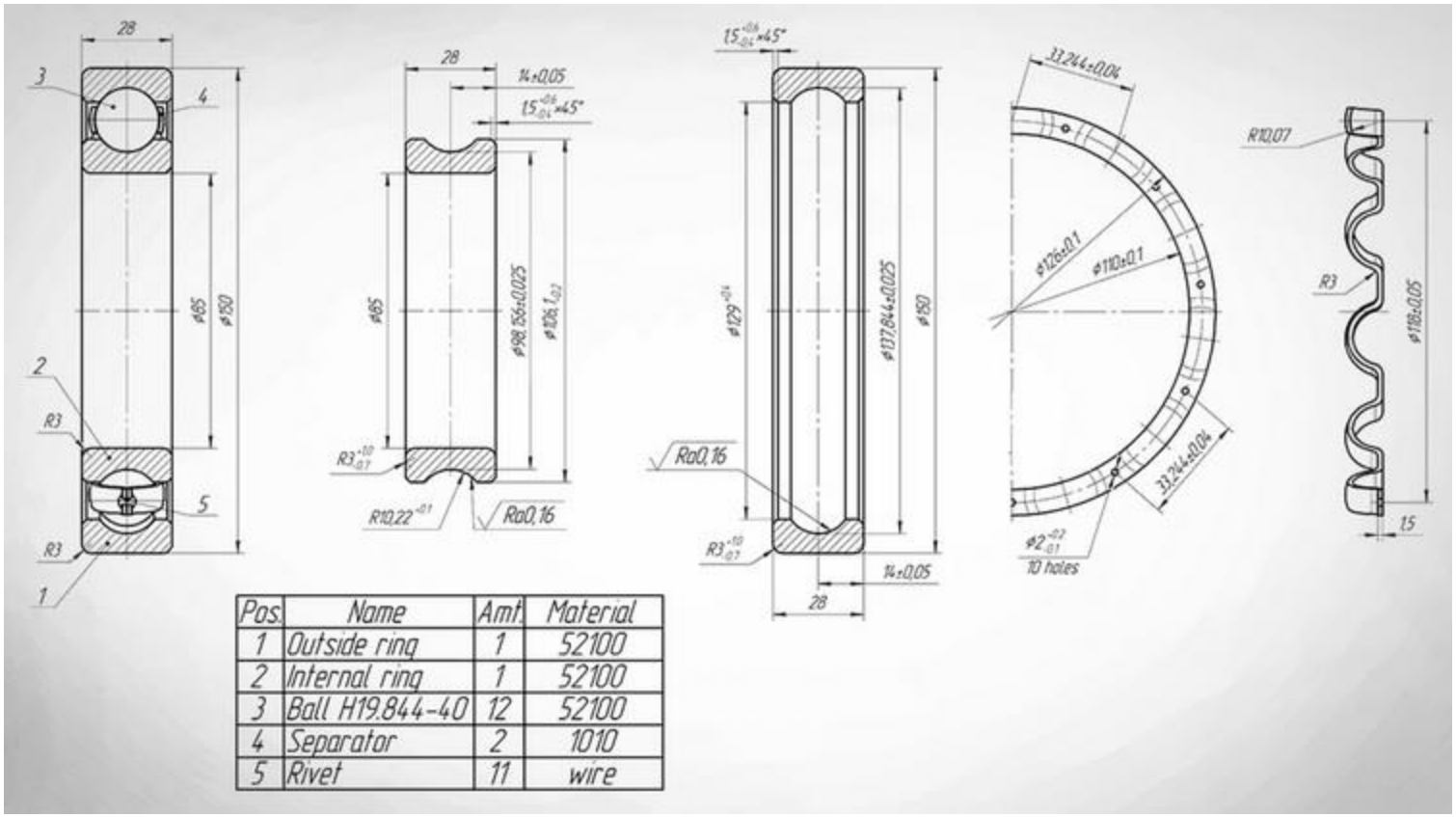
Applications (Engineering Drawing)



Applications (Engineering Drawing)

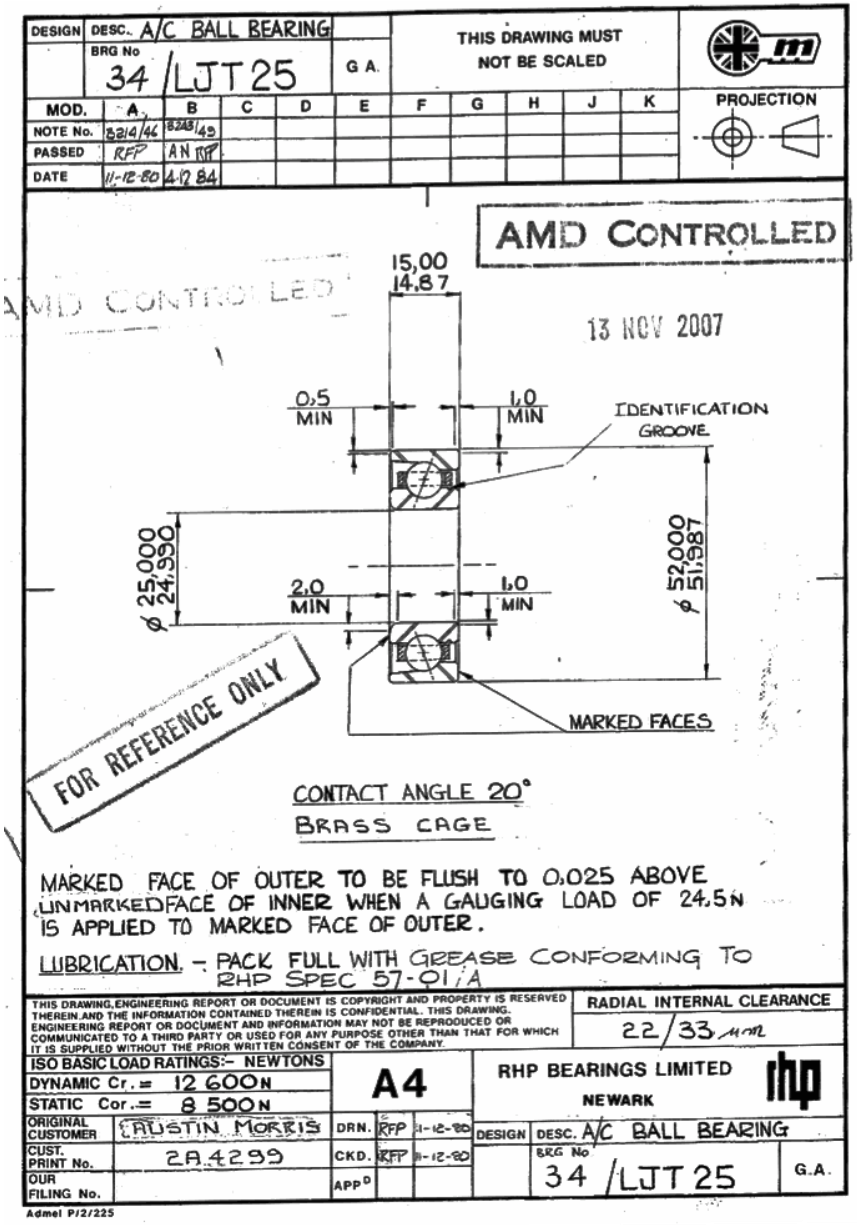


Bearing (Engineering Drawing)



Pos.	Name	Amt.	Material
1	Outside ring	1	52100
2	Internal ring	1	52100
3	Ball H19.844-40	12	52100
4	Separator	2	1010
5	Rivet	11	wire

Bearing (Actual Industry Drawing)



DESIGN	DESC. A/C BALL BEARING										
	BRG No 34 / LJT 25	G A.	THIS DRAWING MUST NOT BE SCALED								
MOD.	A	B	C	D	E	F	G	H	J	K	PROJECTION
NOTE No.	8214/46	8248/49									
PASSED	RFP	AN RFP									
DATE	11-12-80	4-7 84									

AMD CONTROLLED

13 NOV 2007

15,00
14,87

0,5 MIN

1,0 MIN

IDENTIFICATION GROOVE

ø 25,000 24,990

2,0 MIN

1,0 MIN

ø 52,000 51,987

MARKED FACES

FOR REFERENCE ONLY

CONTACT ANGLE 20°

BRASS CAGE

MARKED FACE OF OUTER TO BE FLUSH TO 0,025 ABOVE UNMARKED FACE OF INNER WHEN A GAUGING LOAD OF 24,5N IS APPLIED TO MARKED FACE OF OUTER.

LUBRICATION. - PACK FULL WITH GREASE CONFORMING TO RHP SPEC 57-91/A

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RADIAL INTERNAL CLEARANCE 22/33 mm

ISO BASIC LOAD RATINGS:- NEWTONS
DYNAMIC Cr. = 12 600N
STATIC Cor. = 8 500N

A4

RHP BEARINGS LIMITED
NEWARK

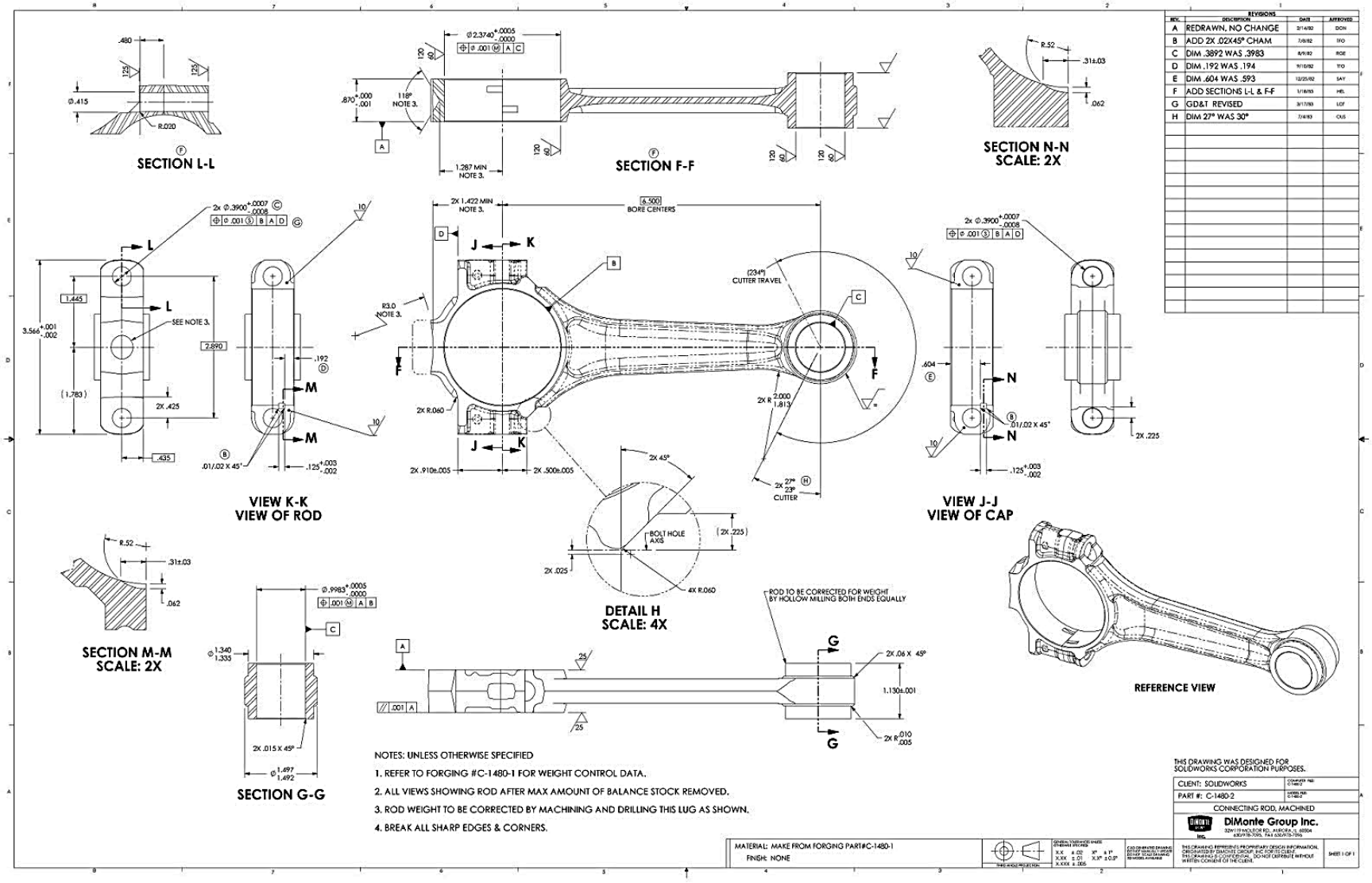
ORIGINAL CUSTOMER	AUSTIN MORRIS	DRN. RFP 1-12-80
CUST. PRINT No.	2A.4299	CKD. RFP 1-12-80
OUR FILING No.		APP'D

DESIGN DESC. A/C BALL BEARING
BRG No 34 / LJT 25 G.A.

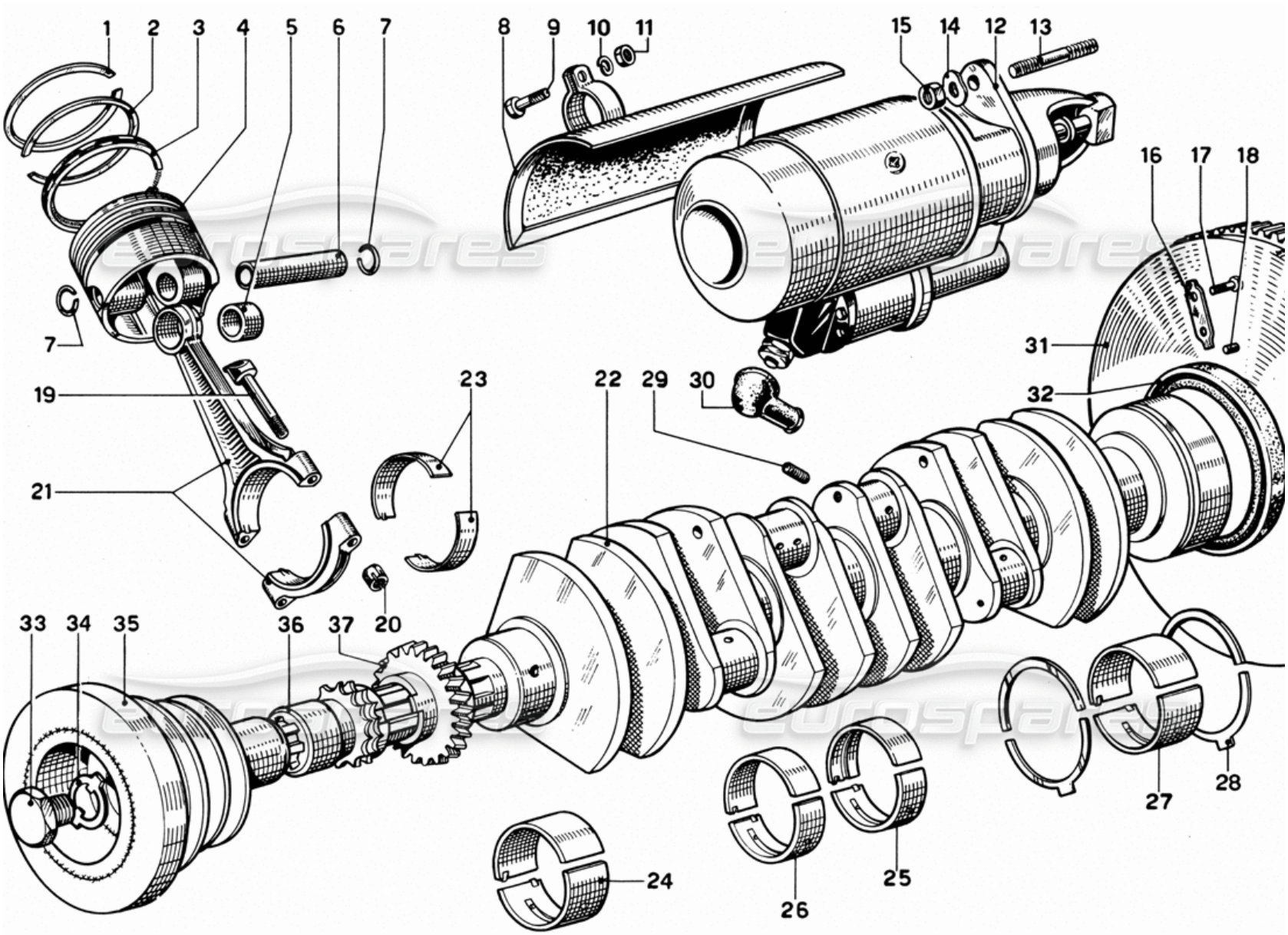
Admel P/2/225

Applications (Engineering Drawing)

Mechanical Engineering

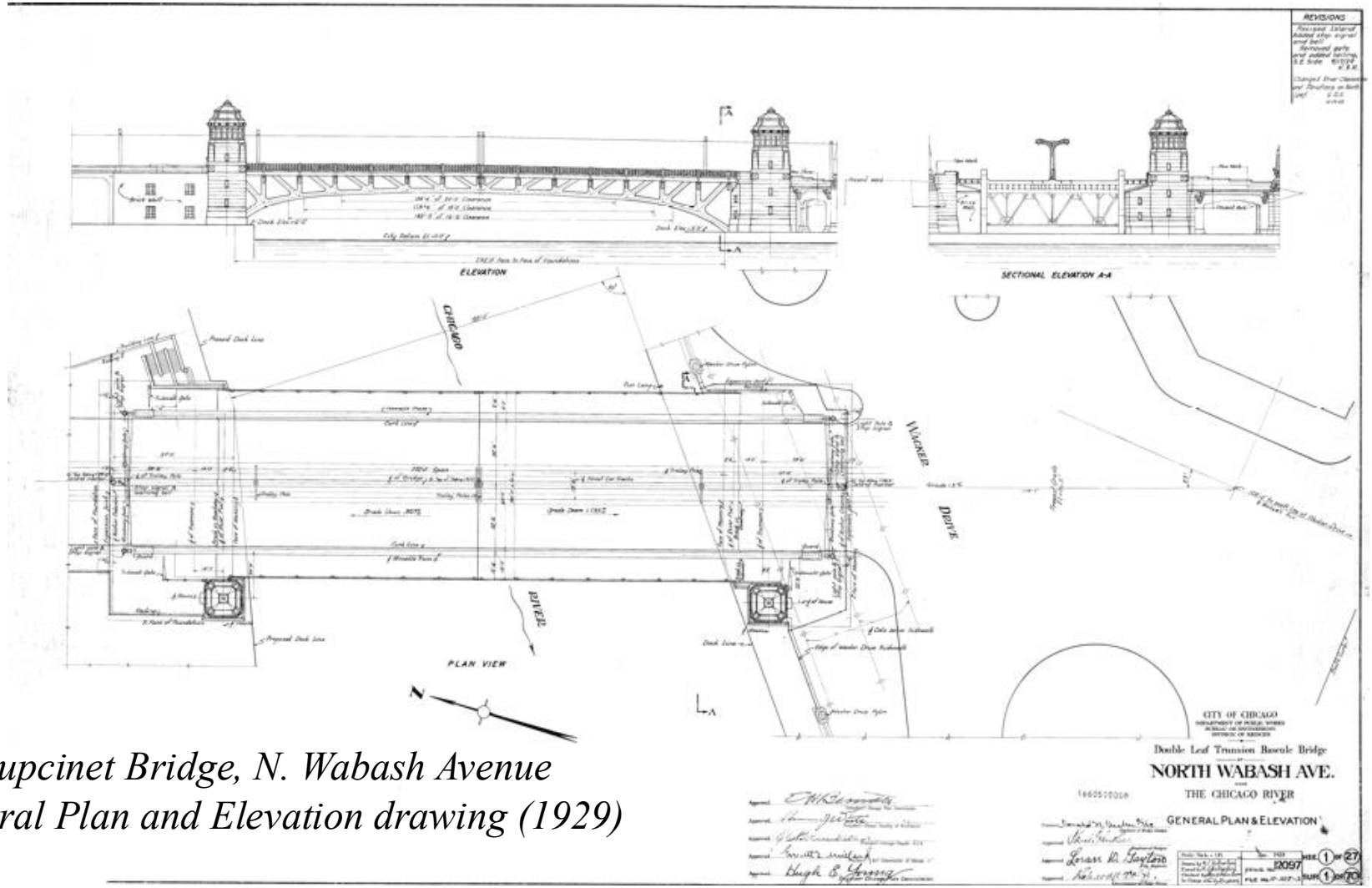


Applications (Engineering Drawing)



Applications (Engineering Drawing)

Civil Engineering (Buildings, Highways, Pipelines, Bridges, Water and Sewage systems.)



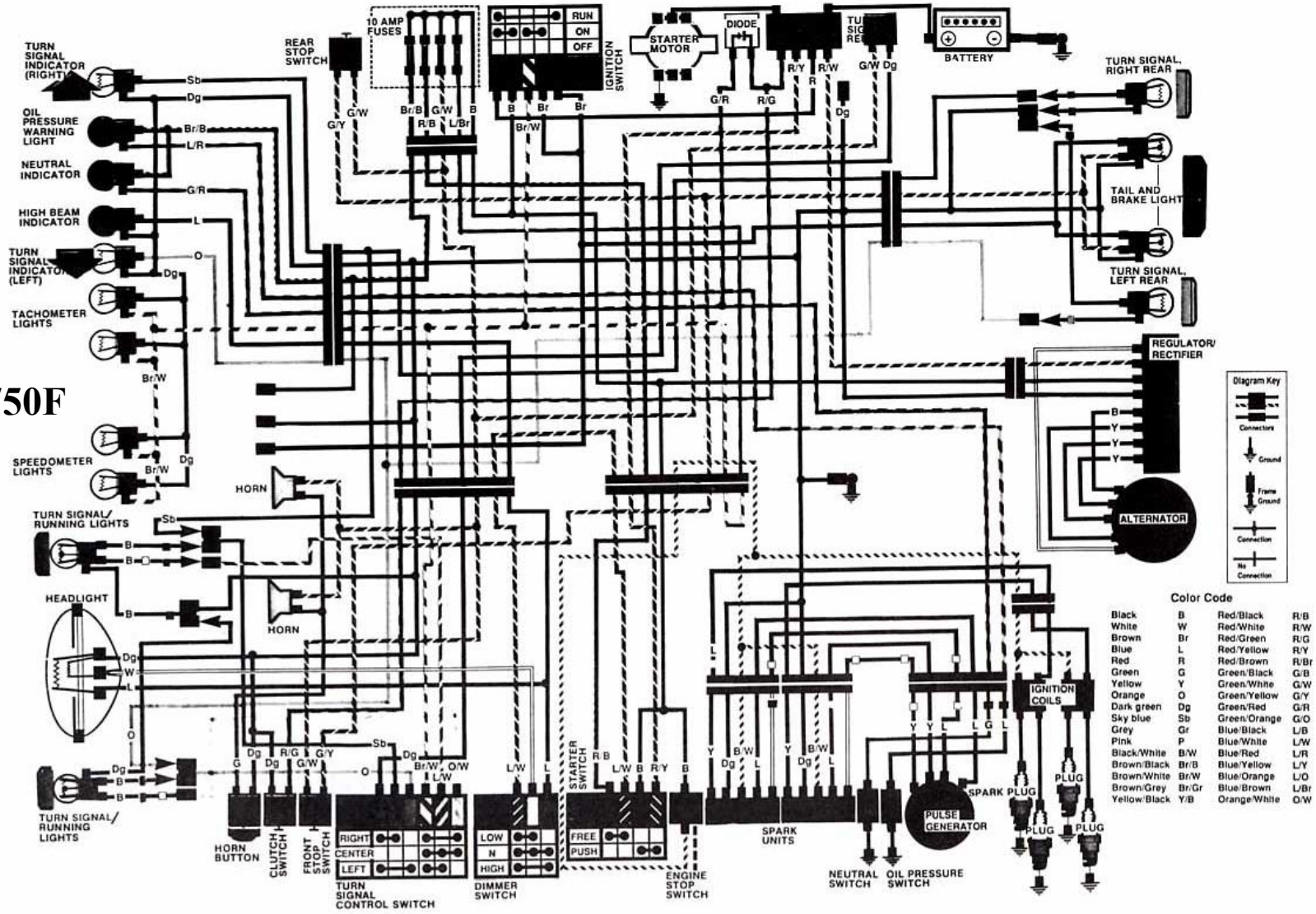
Irv Kupcinet Bridge, N. Wabash Avenue
General Plan and Elevation drawing (1929)

1. Courtesy of the Chicago Department of Transportation - Division of Engineering

Applications (Engineering Drawing)

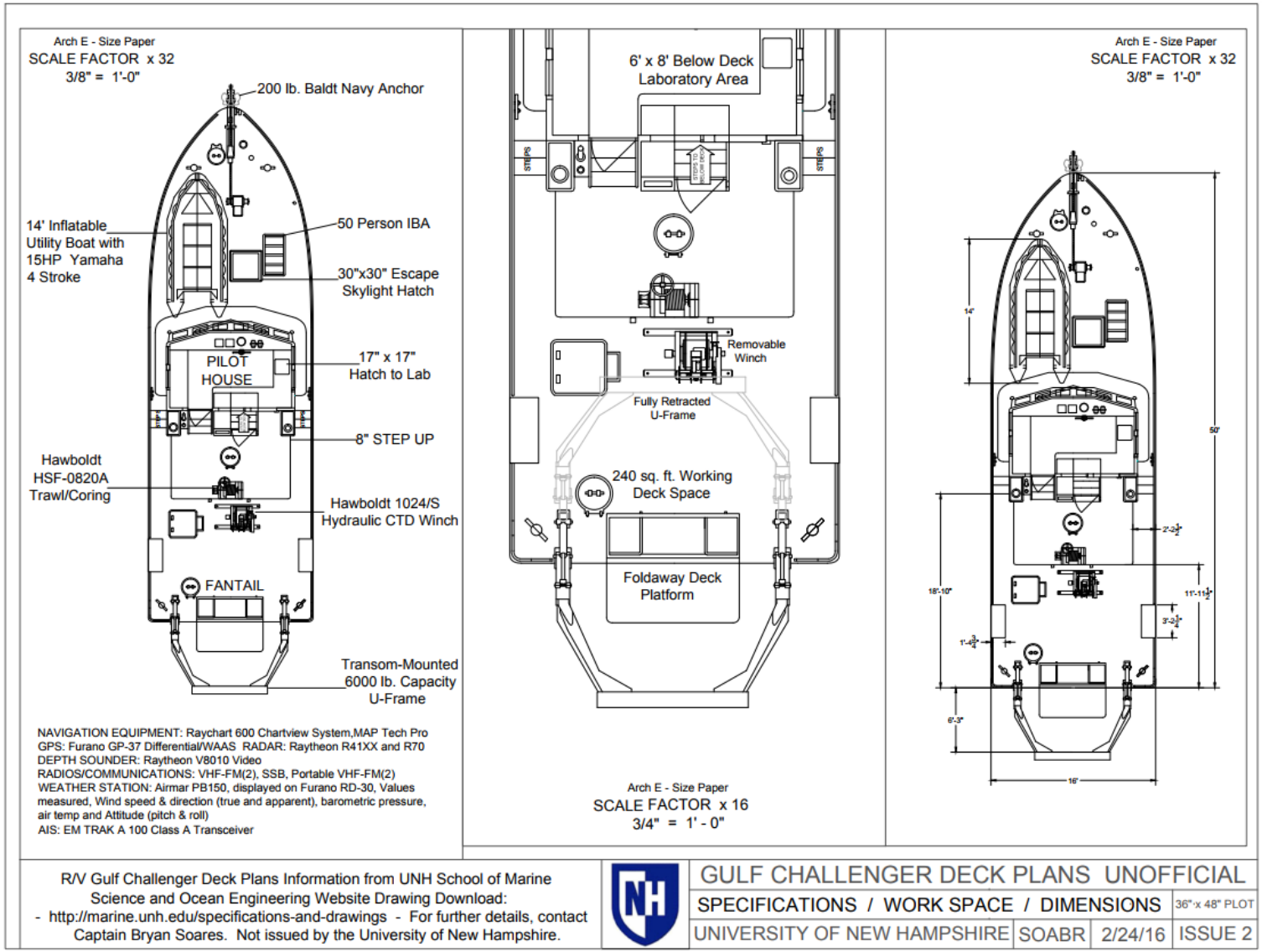
Electrical Engineering (Sensors, Wiring diagrams, Circuit diagrams.)

Honda CB750F



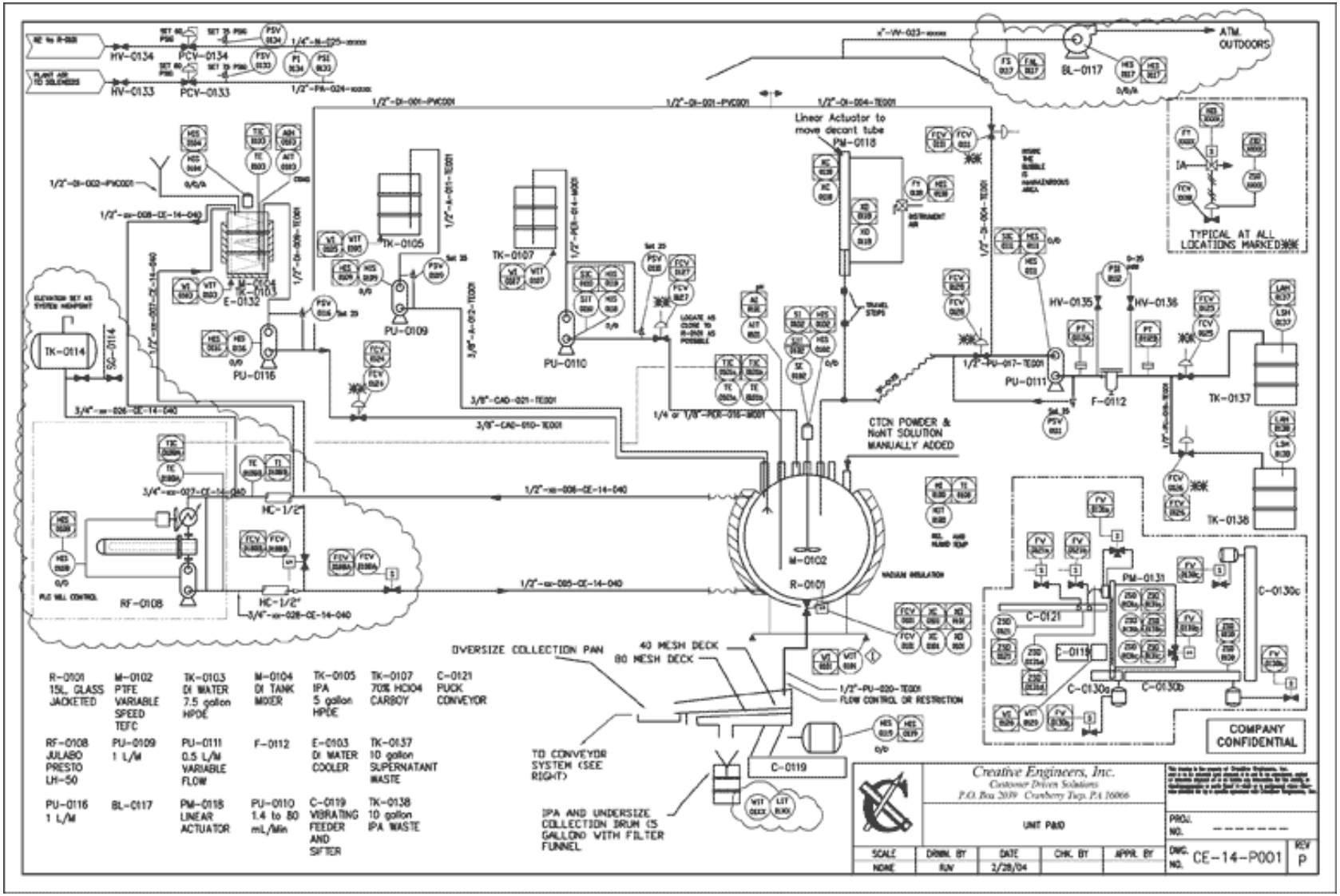
Applications (Engineering Drawing)

Ocean Engineering (Deck plans, other equipment on ship.)



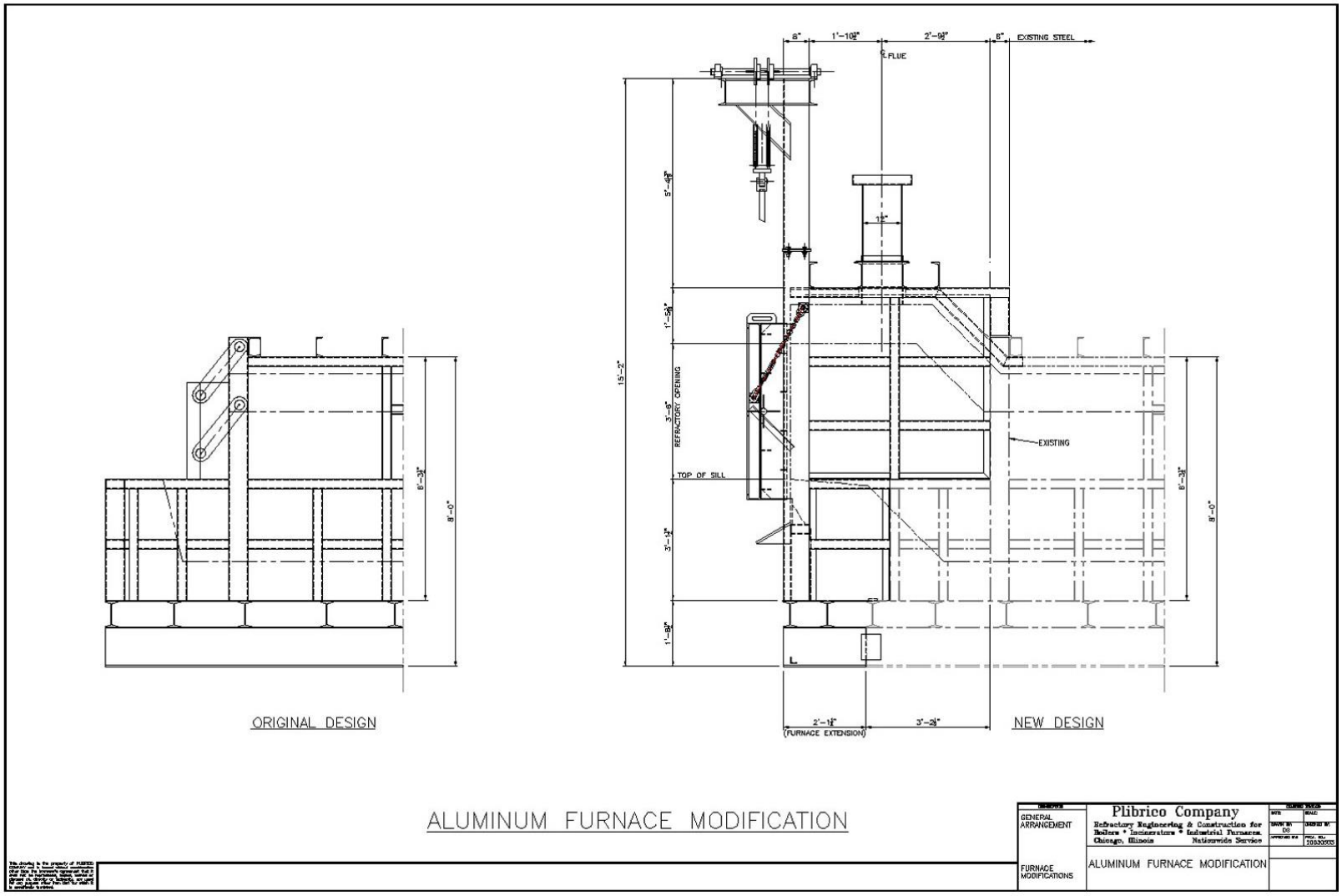
Applications (Engineering Drawing)

Chemical Engineering (Processing plant layout, piping)



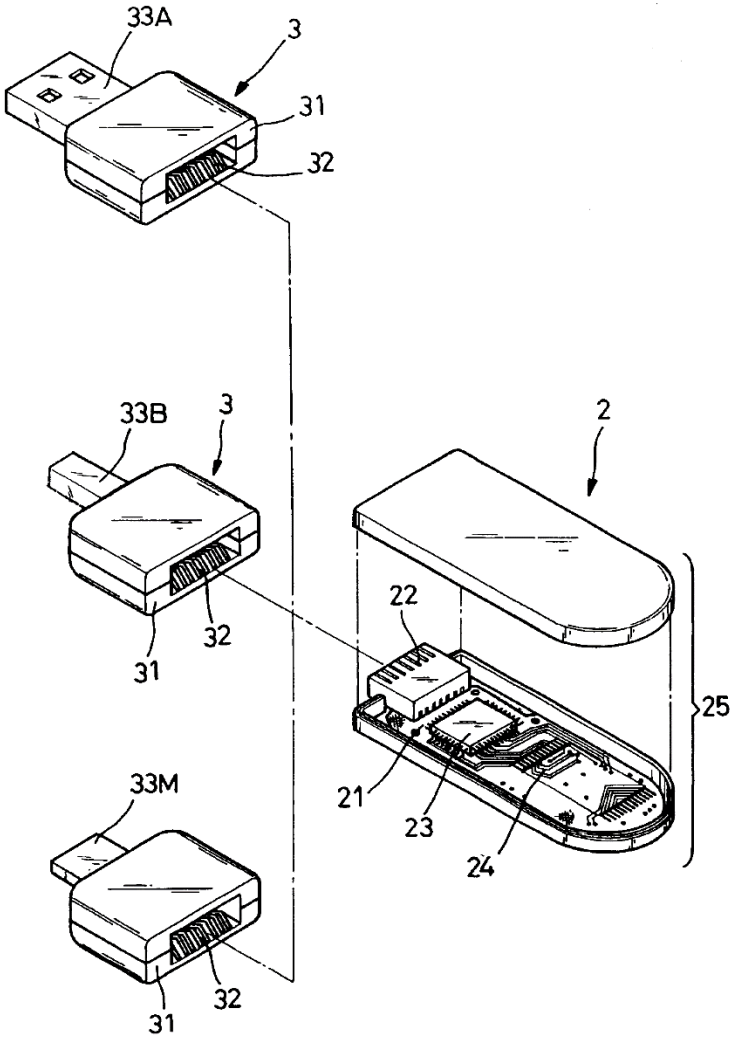
Applications (Engineering Drawing)

Metallurgical and Materials Engineering (Furnace Drawing)



Applications (Engineering Drawing)

Computer Science Engineering (Pen drive patent)



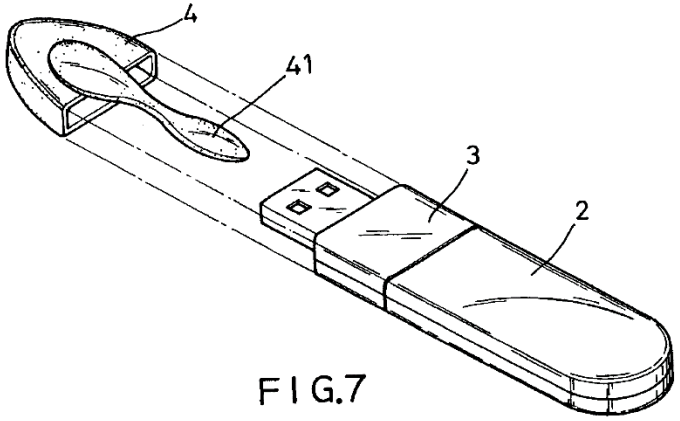


FIG.7

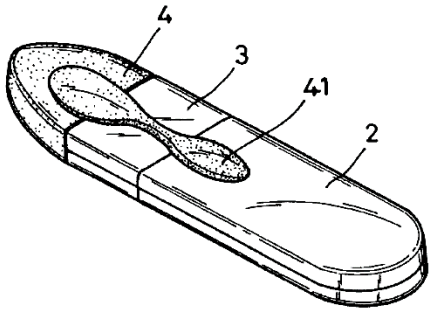


FIG.8

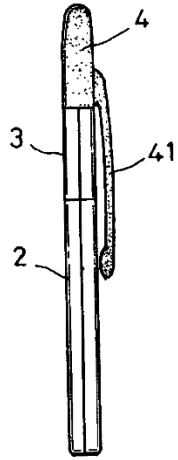


FIG.9

Purpose of Engineering Drawing

1. To convey the geometric content of the part
2. Dimensions
3. Geometric details (hidden lines and surfaces, shapes, section views, etc.)
4. Manufacturing details
5. Tolerances
6. Surface roughness
7. Machining instructions and materials

Paper Size (A series)

1. The size of paper in A series varies from A0 to A10.

2. For the A0 size, the area of the paper is $1m^2$, with the ratio of length to width being $\sqrt{2}:1$.

3. Therefore, the length and the width of A0 sheets are 1189 mm and 841 mm respectively.

4. For every successive paper size A1, A2 etc., the area becomes half of the previous size maintaining the same length to width ratio.

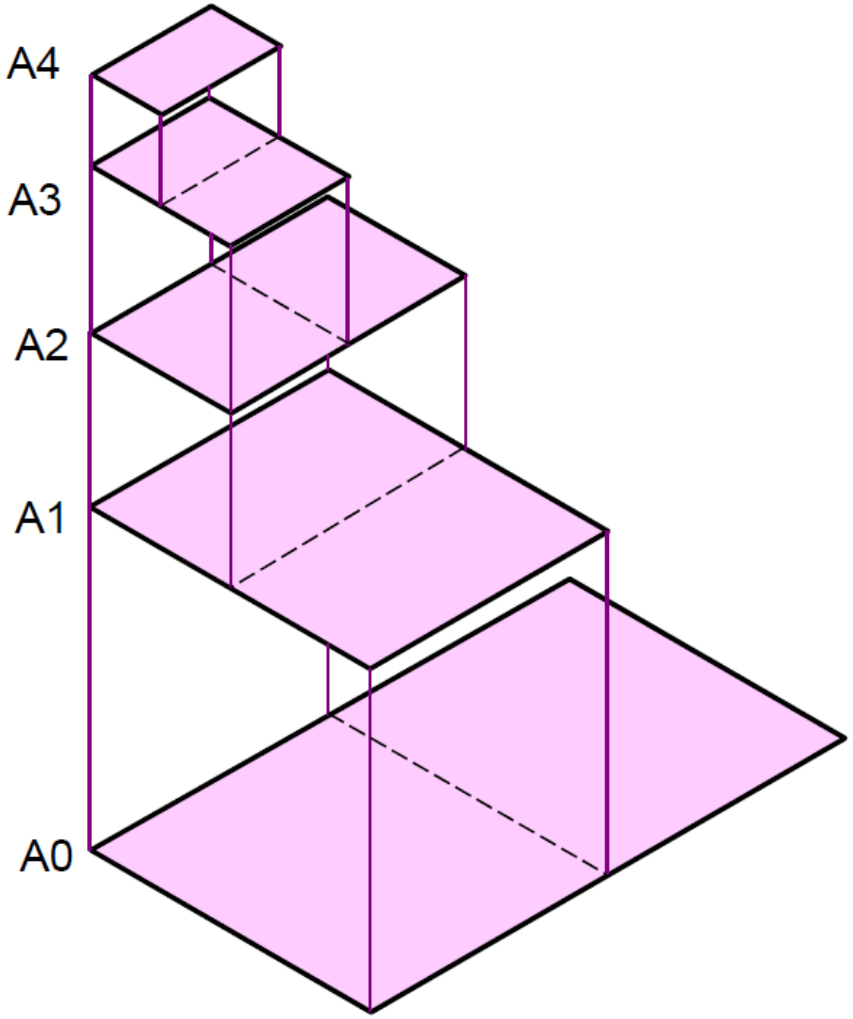
5. Therefore, the length of the succeeding size will be same as width of the preceding size (The length of A1 is same as width of A0).

6. ISO 216:2007 is a document issued by International Organization for standardization for governing various sizes of paper.

Paper Size (A series)

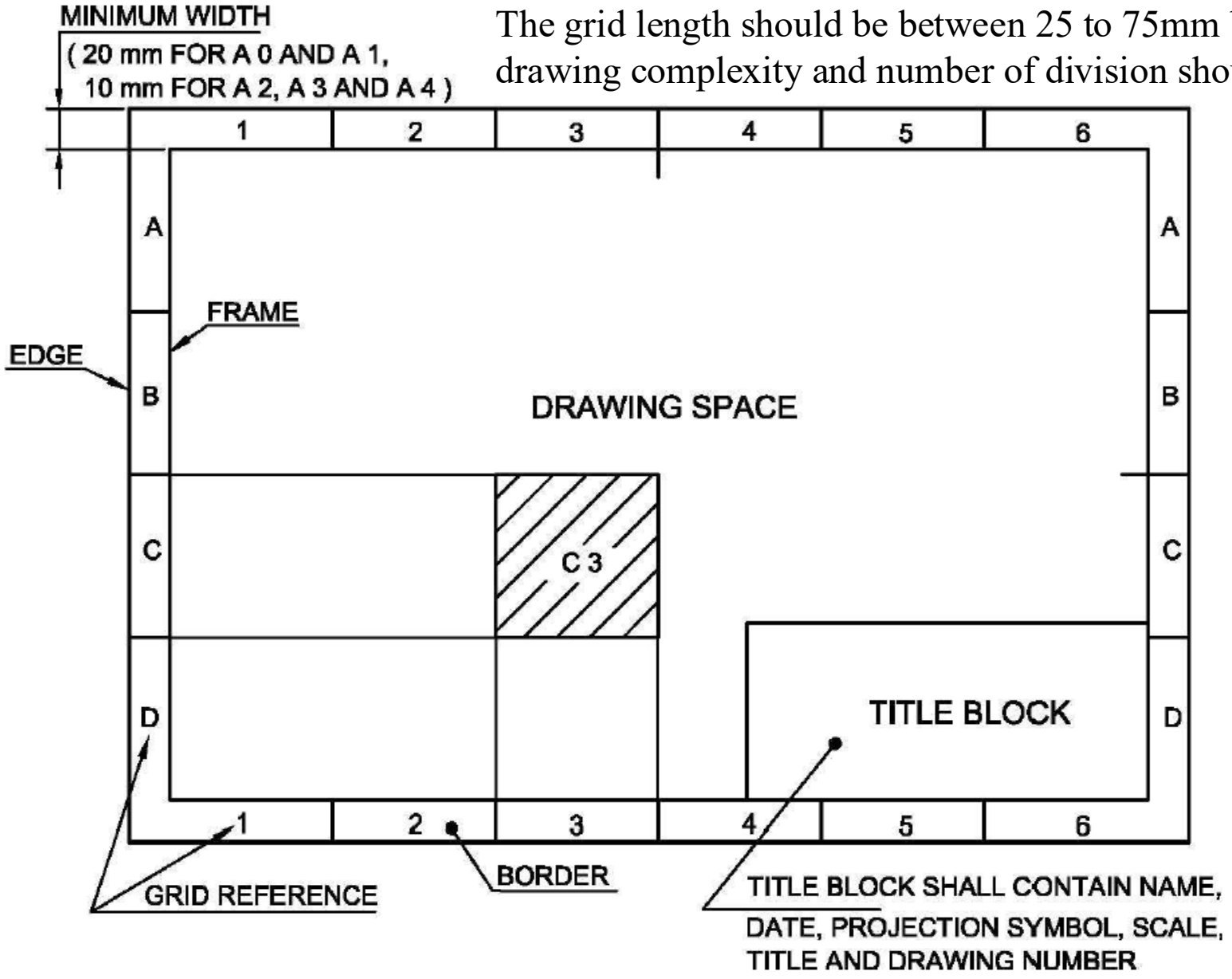
Frequently used sizes:

Paper Size	Width (mm)	Length (mm)	Area (m^2)
A0	841	1189	1
A1	594	841	0.5
A2	420	594	0.25
A3	297	420	0.125
A4	210	297	0.0625



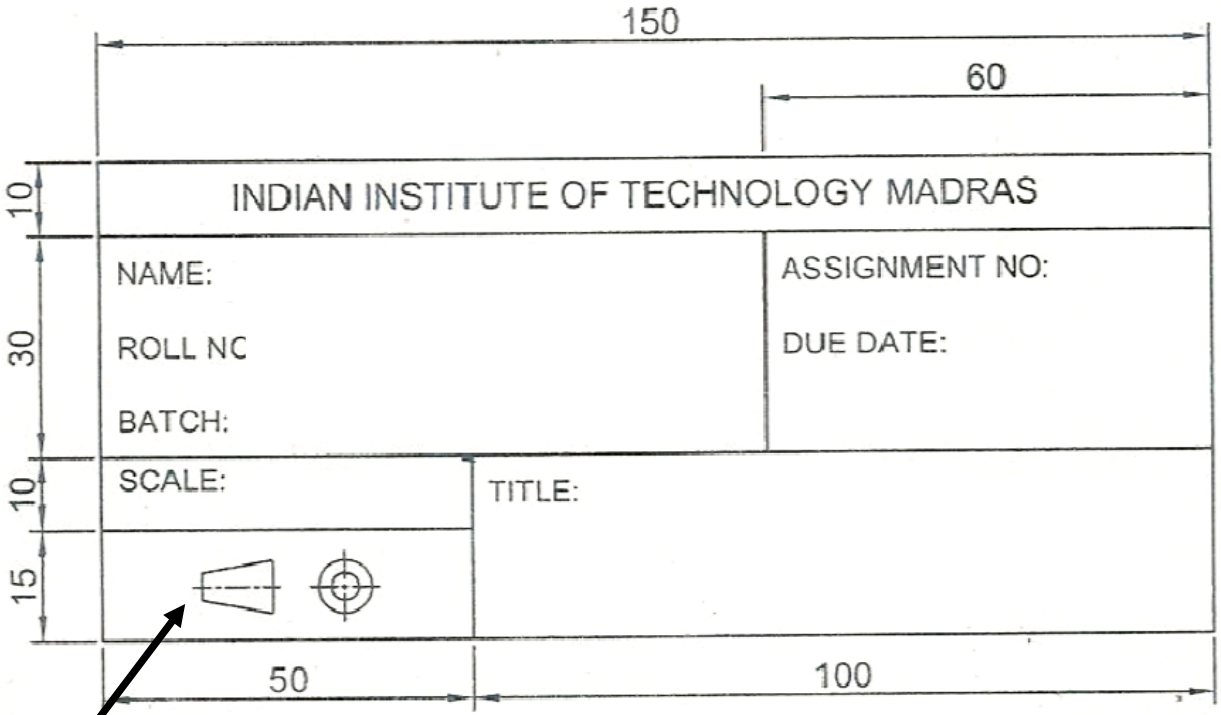
Drawing Sheet Layout

The grid length should be between 25 to 75mm based on the drawing complexity and number of division should be even.



Title Block Layout

Title block shall contain the student's name, date of exercise, projection symbol, scale title, and drawing number.



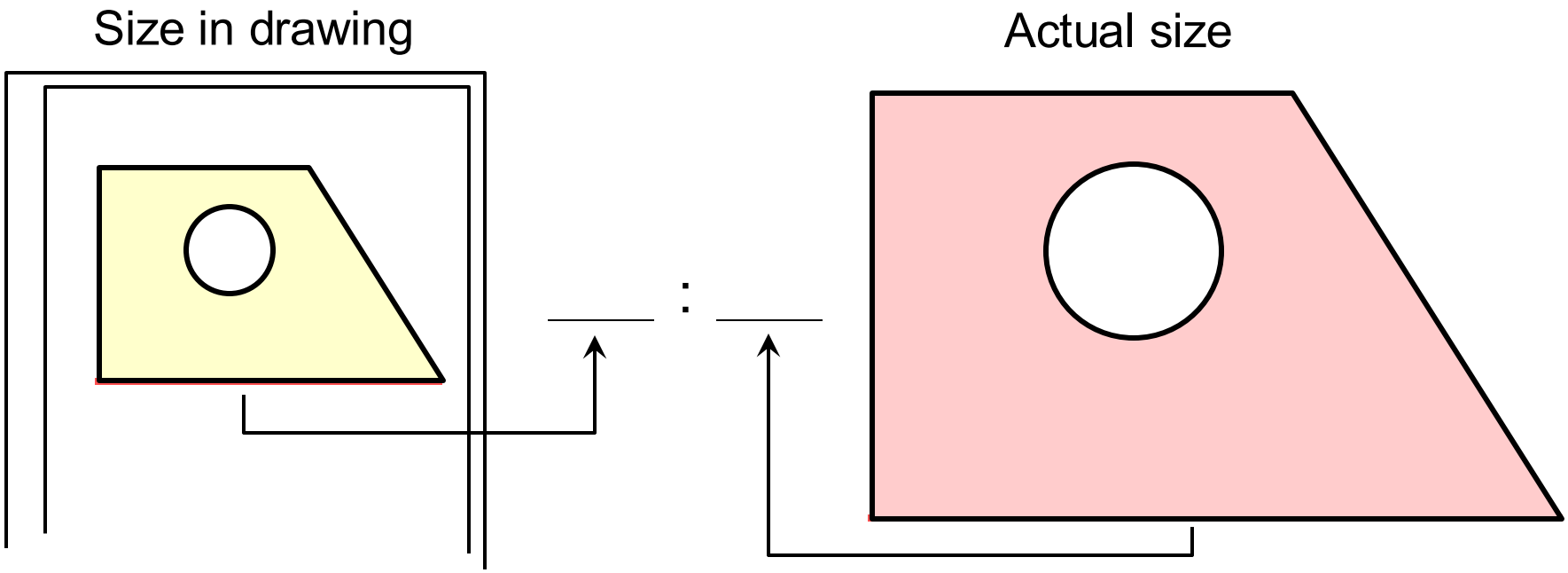
Projection Method Symbol

DETAILS OF TITLE BLOCK
(Suggested for students)

ALL DIMENSIONS IN mm

Drawing Scales

Scale is defined as the ratio of the linear dimensions of the object as represented in a drawing to the actual dimensions.



Dimension numbers shown in the drawing are correspond to "true size" of the object and they are independent of the scale used in creating that drawing.

Drawing Scales



Actual Size

SCALE 1:1 for full size

SCALE **X**:1 for ***enlargement*** scales (X > 1)

SCALE 1:**X** for ***reduction*** scales (X > 1)

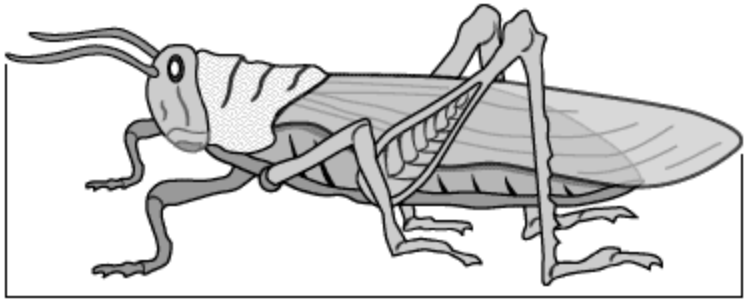


Scale of 1:1



Scale of 1:2

Drawing



Scale of 2:1

Exercise: A mural of a dog was painted on a wall. The enlarged dog was 45 ft. tall. If the average height for this breed of dog is 3 ft., what is the scale factor?

Solution: 15:1

Lettering & Dimensioning

- Text's style on the drawing must have the following 2 properties

Legibility

- Shape
- Space between letters
- Space between words

Uniformity

- Size (or text height)
- line thickness

Examples

ESTIMATE **GOOD**

EstiMaTE Not uniform in style.

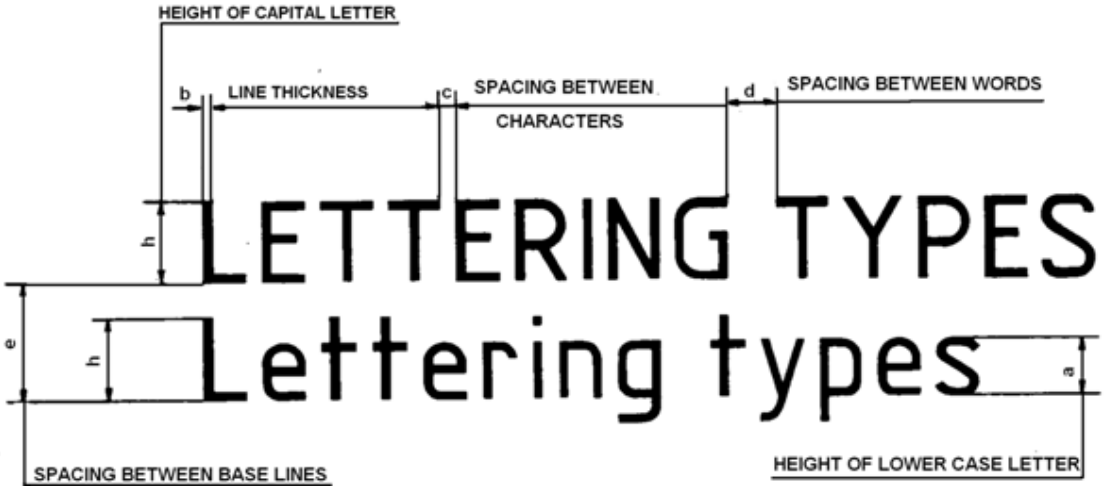
ESTIMATE
ESTIMATE Not uniform in height.

ESTIMATE
ESTIMATE Not uniformly vertical.

ESTIMATE
ESTIMATE Not uniform in thickness of stroke.

ESTIMATE Inappropriate space between letters

Lettering Height



Specifications	Value	Size (mm)					
Capital letter height	h	3.5	5	7	10	14	20
Lowercase letter height	a = (7/10)h	2.5	3.5	5	7	10	14
Thickness of lines	b = (1/10)h	0.35	0.5	0.7	1	1.4	2
Spacing between characters	c = (1/5)h	0.7	1	1.4	2	2.8	4
Min. spacing b/n words	d = (3/5)h	2.1	3	4.2	6	8.4	12
Min. spacing b/n baselines	e = (7/5)h	5	7	10	14	20	28

1. Refer to BIS SP 46(2003) for details 2. http://nptel.ac.in/courses/112103019/2

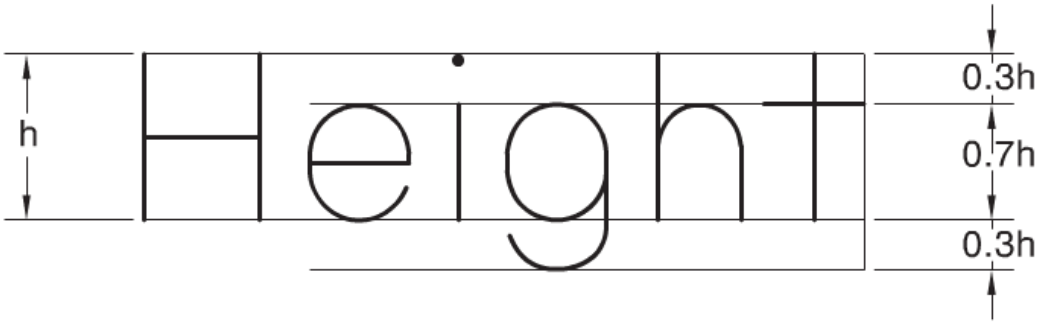
Lettering Height

Lettering type B (Refer to BIS SP46(2003) for details)

Main title: h=7 mm

Subtitles: h=5 mm

Dimensions and notes: h=3.5 mm



Basic Line Types

Types of Lines	Appearance	Name according to application
Continuous thick line		Visible line
Continuous thin line		Dimension line Extension line Leader line
Dash thick line		Hidden line
Chain thin line		Center line

NOTE : We will learn other types of line in practical classes.

Meaning of Lines

Visible lines represent features that can be seen in the current view

Hidden lines represent features that can not be seen in the current view

Center line represents symmetry, path of motion, centers of circles, axis of axi-symmetrical parts

Dimension and Extension lines indicate the sizes and location of features on a drawing

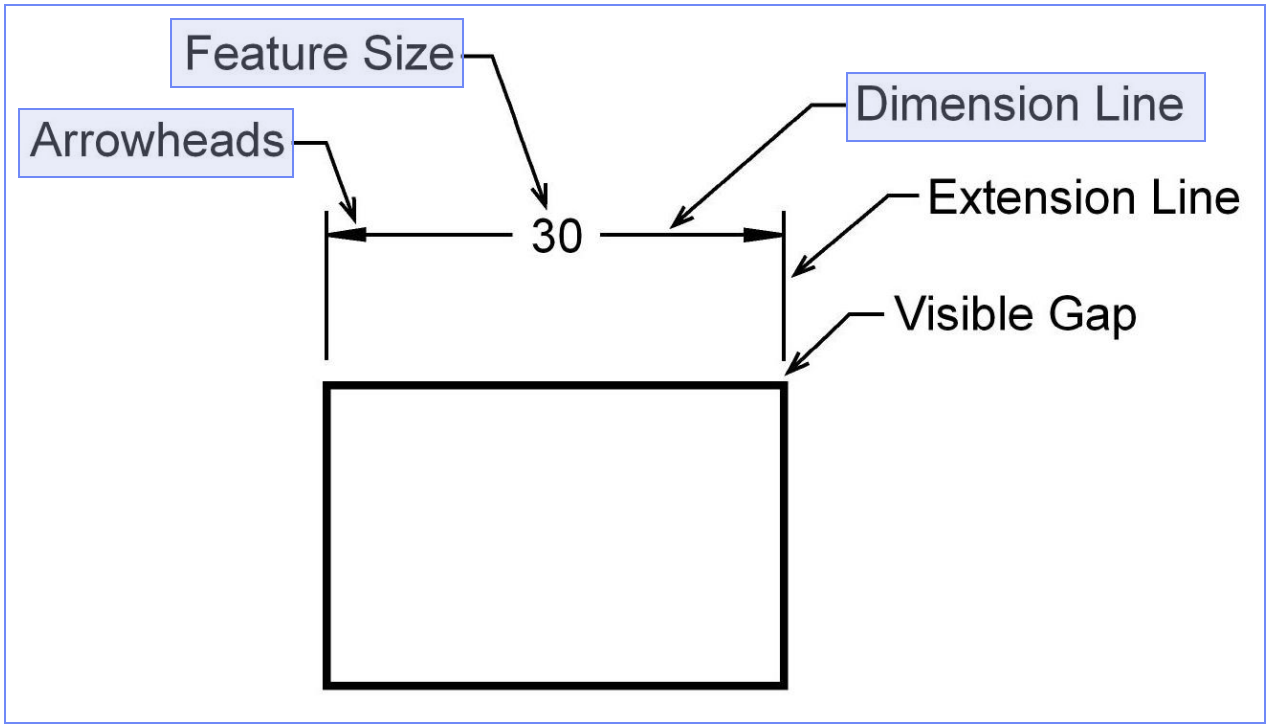
Lines used in Dimensioning

- **Dimensioning requires the use of**
  - → Dimension lines
  - → Extension lines
  - → Leader lines

- All three line types are drawn ***thin*** so that they will not be confused with visible lines.

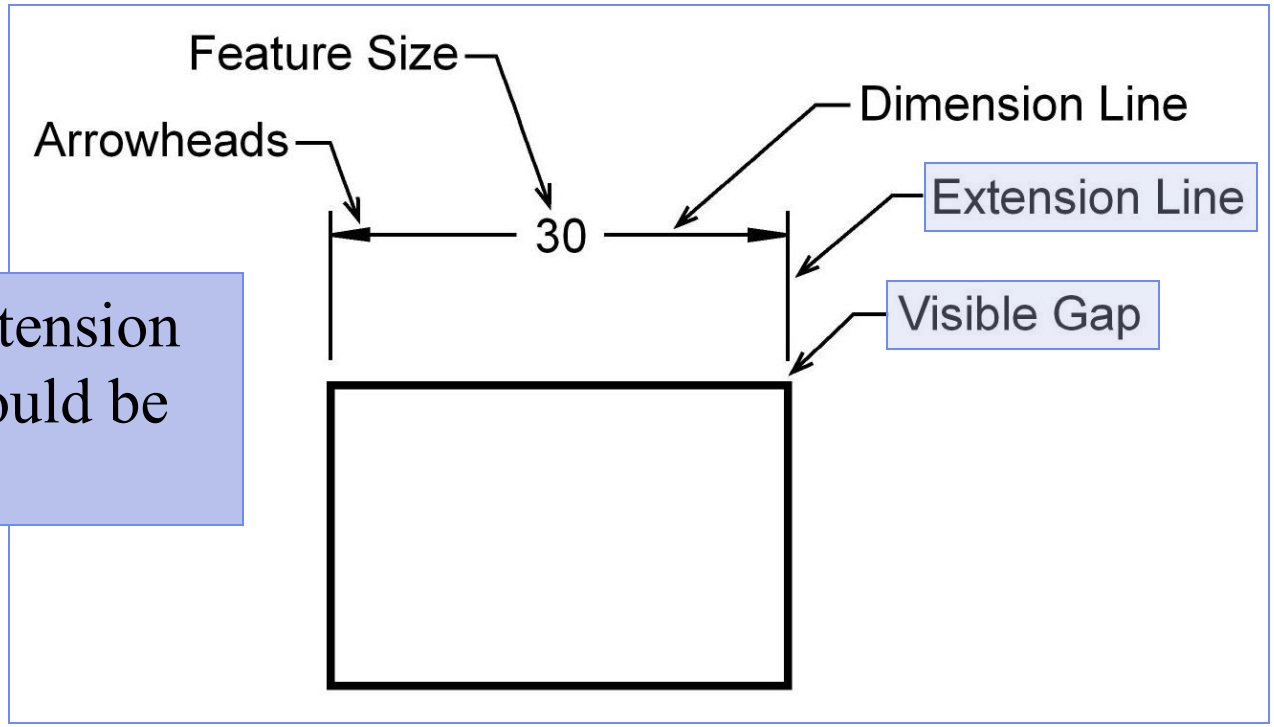
Dimension Line

- **Dimension Line:** A line terminated by arrowheads, which indicates the direction and extent of a dimension.



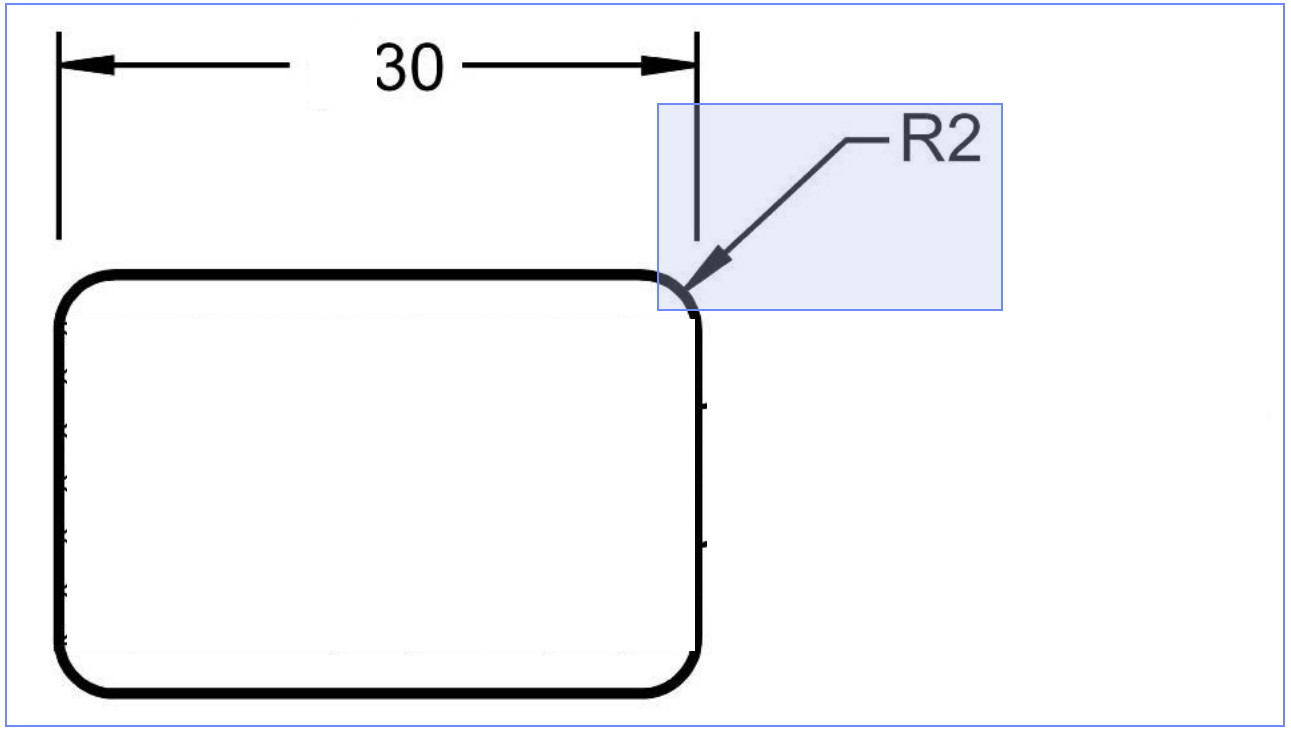
Extension Line

- **Extension line:** An extension line is a thin solid line that extends from a point on the drawing to which the dimension refers.

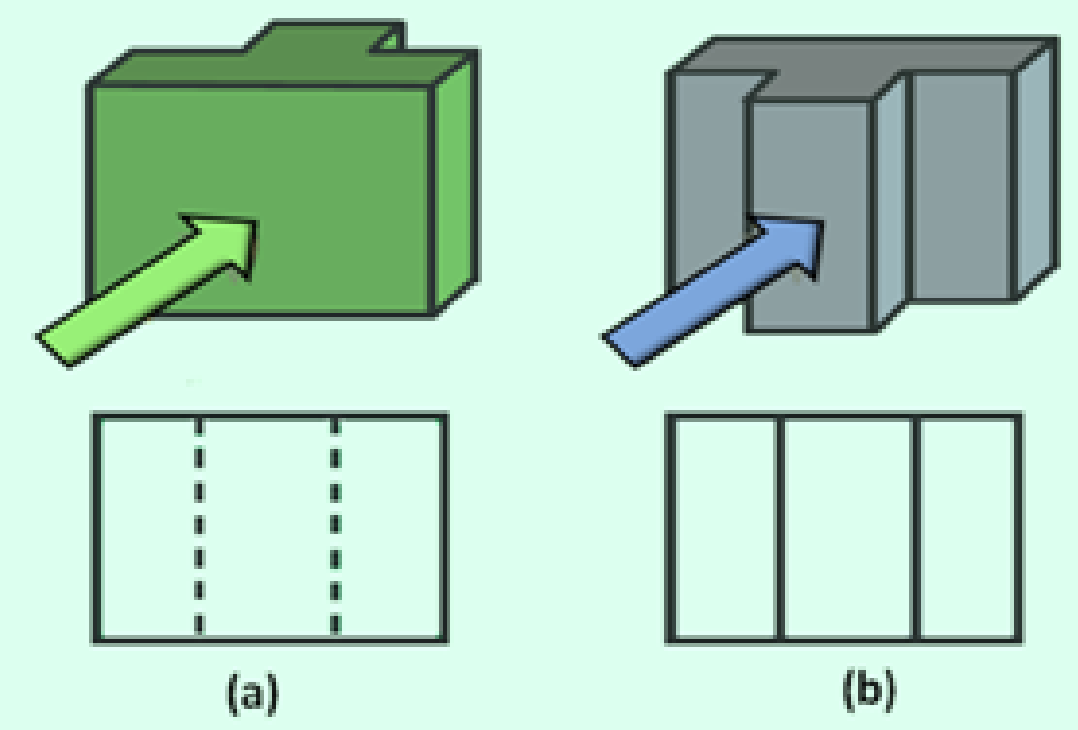


Leader Line

- **Leader Line:** A straight inclined thin solid line that is usually terminated by an arrowhead.

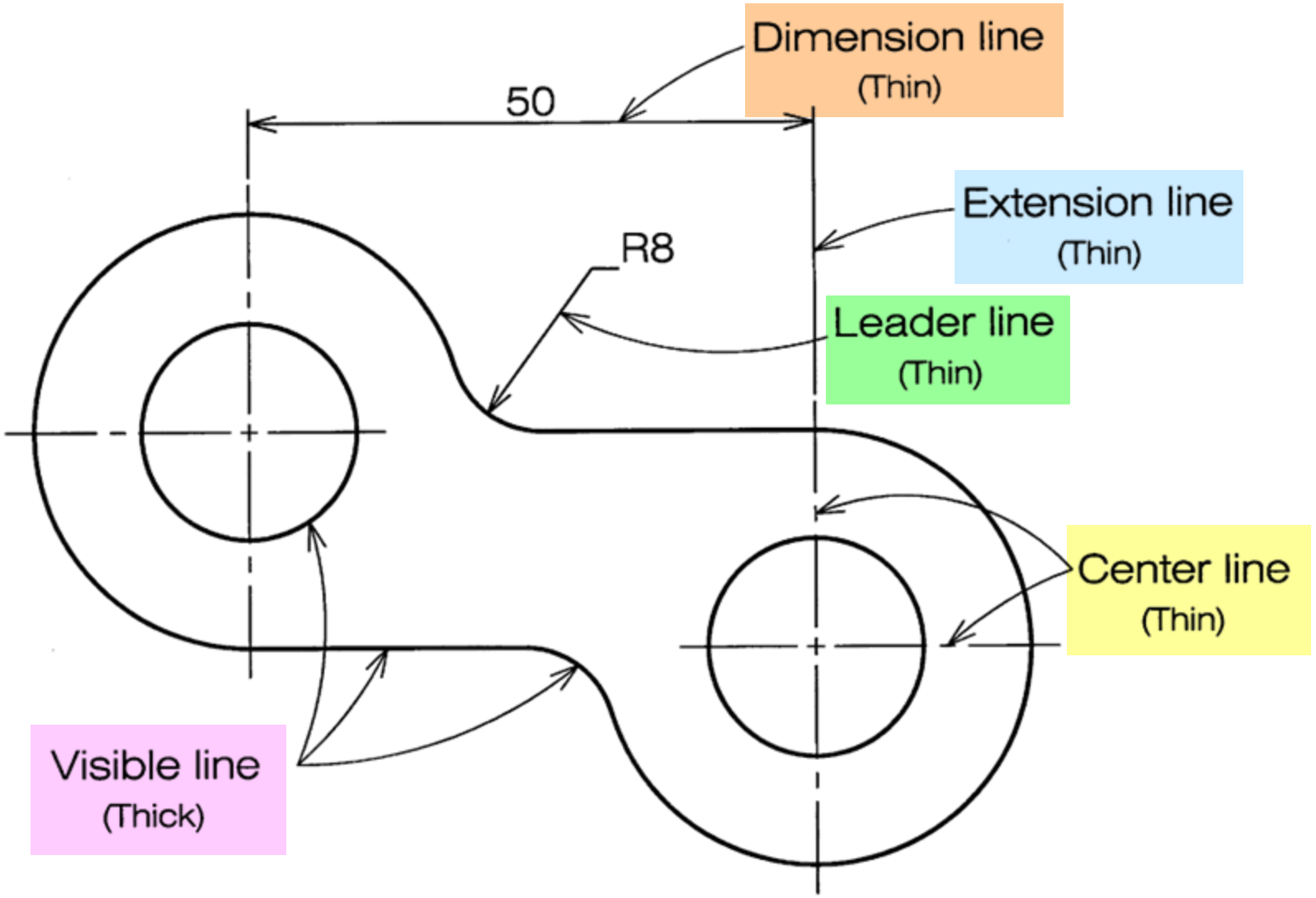


Example



Hidden line Vs Visible line

Example


Dimension line
(Thin)
50
Extension line
(Thin)
R8
Leader line
(Thin)
Center line
(Thin)
Visible line
(Thick)

Guidelines

LINES, LETTERING AND DIMENSIONING

LINE TYPES

Sl. No	LINE TYPE	APPLICATION	PENCIL TYPE
1	Continuous thick ———	Visible edges, Visible outlines	H (Medium grade)
2	Continuous thin ———	Construction lines, Guide lines, Projection lines, Dimension lines, Extension lines, Leader lines and Hatching lines	2H (Hard grade)
3	Dashed thick - - - - -	Hidden outlines, Hidden edges	H
4	Chain thin ———	Center lines, Axes, Lines of symmetry, Trajectories and Pitch circles	2H
5	Continuous thin free hand ~~~	Limits of partial or interrupted views	2H
6		Border lines, Lettering and Free hand skecthing	HB (Soft grade)

SCALES

TYPE	RECOMMENDED		
Enlargement scale	50:1	20:1	10:1
	5:1	2:1	-
Full size	1:1		
Reduction scale	1:2	1:5	1:10
	1:20	1:50	1:100
	1:200	1:500	1:1000
	1:2000	1:5000	1:10000

$$\text{Scale of a drawing / Representative fraction} = \frac{\text{Drawing length of an element}}{\text{Actual length of that element}}$$

(PTO)

Guidelines

LETTERING

Recommended letter sizes

2.5 mm & 3.5 mm - For Dimensioning & Marking

5 mm - For Writing notes & Subtitles

10 mm - For Title of the Drawing



Lettering Types:

1) Vertical single stroke letters

2) Inclined single stroke letters

Shape identification symbol for dimensioning

Ø - Diameter

R - Radius

□ - Square

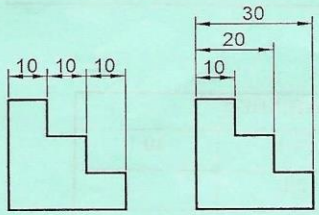
SØ - Spherical diameter

SR - Spherical radius

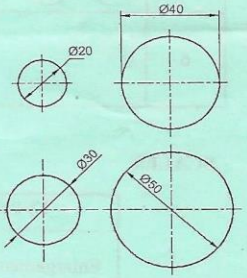
Lettering sizes ratio:

Height of the Upper case letters 'h'	(14/14) h	2.5	3.5	5	7	10	14	20
Height of the Lower case letters 'c'	(10/14) h	-	2.5	3.5	5	7	10	14

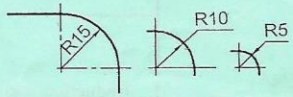
DIMENSIONING METHODS



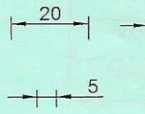
Chain dimensioning Parallel dimensioning



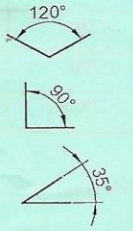
Diameter dimensioning for circles



Radius dimensioning for arcs



Linear dimensioning



Angular dimensioning

Prepared for B.Tech - CS & MM 17, by
S. RAVISUBRAMANIAN, ME
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Dept. of Mechanical Engineering,
IIT Madras, Chennai - 600 036

30/01/2018

Thank you